NBG Intrinsic Automorphic Quantum Axiom System

A Pure Algebraic Origin of Quantumness without External Parameters

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1 Abstract

We present a complete axiomatic foundation for quantum theory within the framework of von Neumann–Bernays–Gödel (NBG) class theory. Quantumness is not imposed through a fixed Planck constant or measurement postulate, but emerges from the breakdown of recursive rigidity across all ordinal levels of class Fourier transforms acting on automorphic forms. The Cauchy–Schwarz equality is identified with global recursive compatibility, forcing the commutator of basic variables to vanish and the spectrum to be purely point-like (compact). A single failure of recursive proportionality at any ordinal level immediately triggers strict inequality, a non-zero commutator, continuous spectrum, and non-compactness. The algebraic Planck constant is geometrically identified as a running first Chern class of the automorphic line bundle, decomposing into a bare integer topological invariant and a scale-dependent vacuum fluctuation correction, directly matching the QCD topological susceptibility and chiral symmetry breaking order parameter. The system admits no intermediate states; the classical/quantum dichotomy is logically deduced from the ordinal recursion law.

2 Axiom 0: Base Space and Objects

Let Γ be a class discrete group in NBG, and X a class symmetric space.

The quantum state space is defined as the generalized automorphic space:

$$\mathcal{A}ut(\Gamma\backslash X, k)$$

The intrinsic variable classes $\mathfrak{X}, \mathfrak{Y}$ are automorphic forms of weight k, satisfying

$$\mathfrak{X}(\gamma z) = j(\gamma, z)^k \mathfrak{X}(z), \quad \forall \gamma \in \Gamma.$$

They generate an operator ring $\mathcal{A}$. The fundamental bilinear inner product on classes is defined by class summation $\langle \cdot, \cdot \rangle := \sum$, with induced norm $\| \cdot \|$.

Remark. The class summation is well-defined on automorphic classes. In Case A (compact operator, pure point spectrum) the sum converges countably; in Case B we introduce a scale-weighted inner product $\langle \cdot, \cdot \rangle_\mu$ to regularize the long-range divergent class sum.

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3 Axiom I: NBG Recursive Cauchy Rigidity Criterion (Fundamental Law)

3.1 Cauchy Inequality on Class Space

$$\|\mathfrak{X}\|^2 \|\mathfrak{Y}\|^2 \geq |\langle \mathfrak{X}, \mathfrak{Y} \rangle|^2$$

3.2 Definition of Recursive Equality

Let $\mathcal{F}$ be the class Fourier transform, and $\mathcal{F}^n$ the n-th recursive Fourier map at ordinal level n. For a limit ordinal $\lambda = \sup_{\alpha<\lambda} \alpha$, define the projective limit $\mathcal{F}^\lambda := \varprojlim_{\alpha<\lambda} \mathcal{F}^\alpha$, and assume that this limit exists in the class $\mathcal{A}$ (its strict closure is guaranteed by Supplementary

Axiom V).

The scale-dependent recursive equality is defined as

$$\mathfrak{X} \overset{\mathrm{rec},\mu}{=} \mathfrak{Y} \iff \forall n \in \mathrm{Ord} : \mathcal{F}^n[\mathfrak{X}] = \lambda_n(\mu)\mathcal{F}^n[\mathfrak{Y}],$$

where $\lambda_n(\mu)$ is the linear proportionality coefficient at ordinal level n and energy scale μ.

Core Rule. The Cauchy inequality becomes an equality **iff** the rigid linear dependence holds across all ordinal levels. If there exists any ordinal n where the proportionality breaks, the inequality is strict.

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4 Axiom II: Spectral Rigidity Dichotomy (Core Judgement)

Based on Axiom I, the system admits a strict dichotomy with no intermediate regime.

4.1 Case A: Recursive Rigidity (Cauchy Equality)

$$\boxed{\mathfrak{X} \overset{\mathrm{rec},\mu}{=} \mathfrak{Y} \iff [\mathfrak{X}, \mathfrak{Y}] = 0 \iff \mathcal{F} \text{ recursively invertible} \iff \sigma(\mathfrak{X}) \text{ compact} \iff \text{self-adjoint compact op}}$$

1. The spectral projection $E_D([R, \infty))$ tends to a compact set as $R \to \infty$.
2. The system is in a classically compatible state; the intrinsic variables are fully resolvable and commute.

4.2 Case B: Recursive Breakdown (Strict Cauchy Inequality)

$$\boxed{\mathfrak{X} \overset{\mathrm{rec},\mu}{\neq} \mathfrak{Y} \iff [\mathfrak{X}, \mathfrak{Y}] = i \cdot \mathbf{C}(\mu)\ (\mathbf{C}(\mu) \neq 0) \iff \mathcal{F} \text{ recursively non-invertible (non-trivial kernel)} \iff \sigma}$$

1. The evolution operator escapes the compact ideal; the spectrum is unbounded and contains a continuous essential spectrum.

2. The system is an intrinsically non-commutative quantum state. Uncertainty originates purely from the geometric structure of automorphic classes, without any measurement-disturbance hypothesis.

4.3 Recursive Rigidity Theorem

- Recursive compatibility + Cauchy equality $\implies$ spectrum is compact with only discrete point spectrum.
- Existence of a continuous essential spectrum $\implies$ the recursive Fourier map has a non-zero kernel $\implies$ recursive relation breaks down.

There exists **no** system simultaneously satisfying "global recursive compatibility, Cauchy equality, and non-compact operator".

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5 Axiom III: Algebraic Topological Running Scalar $\mathbf{C}(\mu)$

5.1 1. Scale-Dependent Topological Average

The scale-dependent topological average is defined as the expectation of the commutator:

$$\mathbf{C}(\mu) := \langle [\mathfrak{X}, \mathfrak{Y}] \rangle_\mu = c_1^{\text{bare}} + \delta c_1(\mu)$$

where $\mathcal{L}_{\text{aut}}$ is the automorphic line bundle associated with the weight-k automorphic forms. To ensure the global periodicity constraint (Appendix A.1), the decomposition is subject to the irreducibility condition:

$$c_1^{\text{bare}} \in \mathbb{Z}, \quad \delta c_1(\mu) \in \mathbb{R}, \quad \text{and} \quad \delta c_1(\mu) \equiv 0 \pmod{\mathbb{Z}} \iff \text{Case A is restored}$$

5.2 2. Topological Decomposition Theorem

$$c_1^{\mathrm{eff}}(\mu) = c_1^{\mathrm{bare}} + \delta c_1(\mu)$$

- $c_1^{\mathrm{bare}} \in \mathbb{Z}$: the bare first Chern class, an absolute integer topological invariant independent of scale;
- $\delta c_1(\mu)$: the running correction induced by ordinal recursive layering and vacuum topological fluctuations, equivalent to the QCD topological susceptibility $\chi_{\mathrm{top}}(\mu)$.

5.3 3. Gauge-Field Physical Correspondence

$\mathbf{C}(\mu)$ acts as the order parameter for the breaking strength of QCD chiral symmetry and the axial $U(1)_A$ symmetry:

1. $\mathbf{C}(\mu) \to 0$: chiral/axial symmetry restored phase, commutator vanishes, spectrum returns to compactness;
2. $\mathbf{C}(\mu) > 0$: spontaneously broken, confining phase, significant non-commutative effects;
3. The QCD θ-vacuum phase linearly superimposes onto the correction $\delta c_1(\mu)$, endowing the effective topological scalar with a periodic topological structure that naturally accommodates strong CP effects.

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6 Axiom IV: Spectral Triple Scaling Law (Adapted to Running $\mathbf{C}(\mu)$)

Based on the Dixmier spectral triple $(\mathcal{A}, \mathcal{H}, D)$ and the non-commutative central limit theorem, four characteristic spectral-geometry parameters are defined:

Parameter	Definition	Physical/Geometric Interpretation
r^*	$\sup\{r : \mathrm{Tr}_\omega(aD^{-r}) < \infty\}$	Short-range repulsion: critical exponent where the Dixmier trace becomes definable

Parameter	Definition	Physical/Geometric Interpretation
r^+	$\inf\{R : E_D([R,\infty))$ non-compact$\} \sim \mu^{-1}$	Long-range spectral cutoff: energy scale from which spectral projections lose compactness
d_x	$\Re(d_{\text{spectral}})$	Transverse fractal spectral dimension (real part)
d_y	$\Im(d_{\text{spectral}})$	Longitudinal oscillatory spectral dimension (imaginary part)

6.1 Unified Limit Relation for the Topological Scalar

$$\mathbf{C}(\mu) = \lim_{\substack{r^* \to 0 \\ r^+ \to \mu^{-1}}} \left[\frac{\mathrm{Vol}_{\text{Dixmier}}(r^*, r^+)}{\mathrm{Vol}_{\text{classical}}(d_x, d_y)} \right]$$

1. $\mathrm{Vol}_{\text{Dixmier}}$: non-commutative geometric measure volume; $\mathrm{Vol}_{\text{classical}}$: classical automorphic measure volume.
2. The limit exists and is non-zero **iff** the system belongs to Case B (non-compact spectrum, quantum non-commutative phase); in Case A the limit is identically zero.
3. The infrared cutoff satisfies $r^+ \sim 1/\mu$, making the limit vary continuously with energy scale μ and generating the renormalisation flow $\mu \mapsto \mathbf{C}(\mu)$.

7 Supplementary Axiom V (Ordinal Closure and Projective Limit Completeness, Optional)

Motivation: Axiom I involves recursive maps $\mathcal{F}^n$ for all ordinals $n \in \mathrm{Ord}$. The definition at a limit ordinal λ depends on the inverse limit $\varprojlim_{\alpha<\lambda} \mathcal{F}^\alpha$. This axiom ensures that this object is well-defined and unique within the operator class $\mathcal{A}$. As an optional supplementary clause, this axiom does not participate in the dichotomy judgement of Axiom II; it merely ensures the set-theoretic legitimacy of the transfinite recursion steps.

Axiom V.1 (Projective Closure of Class $\mathcal{A}$)

Let λ be any limit ordinal. If $\{A_\alpha\}_{\alpha<\lambda}$ is any inverse system in $\mathcal{A}$ whose transition morphisms $\pi_{\alpha,\beta} : A_\beta \to A_\alpha$ $(\alpha \le \beta < \lambda)$ are continuous in the strong class topology, then its projective limit:

$$\varprojlim_{\alpha<\lambda} A_\alpha := \left\{ (a_\alpha)_{\alpha<\lambda} \in \prod_{\alpha<\lambda} A_\alpha \;\middle|\; \forall\alpha \le \beta < \lambda,\ \pi_{\alpha,\beta}(a_\beta) = a_\alpha \right\}$$

exists, is unique, and belongs to $\mathcal{A}$. That is, $\mathcal{A}$ is closed under projective limits of arbitrary length.

Axiom V.2 (Limit Compatibility of the Recursive Fourier Transform)

For a limit ordinal $\lambda = \sup_{\alpha<\lambda} \alpha$, define:

$$\mathcal{F}^\lambda := \varprojlim_{\alpha<\lambda} \mathcal{F}^\alpha$$

and require that this limit is compatible with the recursive maps at all lower levels, i.e., for all $\beta < \lambda$:

$$\pi_\beta \circ \mathcal{F}^\lambda = \mathcal{F}^\beta \circ \pi_\beta$$

where $\pi_\beta : \mathcal{F}^\lambda \to \mathcal{F}^\beta$ is the natural projection. This condition ensures that the judgement at $n = \lambda$ in Axiom I reduces to the collective behaviour of all levels $\alpha < \lambda$, introducing no extra degrees of freedom.

Axiom V.3 (Ultrafilter Completeness, Optional Strengthening)

If $\mathcal{A}$ is regarded as a class topological ring in NBG, we additionally require that $\mathcal{A}$ remains closed under inverse limits along the regular filter $\mathcal{U}$ containing all terminal segments $[\alpha, \lambda)$. This condition is equivalent to asserting the existence of a unique ultrafilter-compatible topology on $\mathcal{A}$ such that all projective limits at limit ordinals coincide with the categorical notion of limits.

Supplementary Remark: If only finite or countable recursion ($n \in \mathbb{N}$) is considered, this axiom can be entirely ignored without affecting the argument of the Final Theorem.

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8 Final Theorem (Holographic Compression): NBG Quantum Dichotomy Law

The classical/quantum phase of intrinsic variable classes is determined solely by the equality condition of the recursive Cauchy inequality; the two phases admit no continuous interpolation.

1. **Cauchy equality (Classical Phase)**

$$\iff \text{global recursive rigidity} \iff \text{spectrum compact with only discrete point spectrum} \iff \text{self-a}$$

2. **Strict Cauchy inequality (Quantum Phase)**

$$\iff \exists \text{ an ordinal level with recursive breakdown} \iff \text{spectrum non-compact with continuous esse}$$

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9 Appendix A: Topological Charge Integrality Constraint and Extended Boundary

This appendix clarifies the topological charge integrality constraint enforced by the global periodicity of the complex exponential function, as well as the extended boundary mechanism of this constraint within the recursive framework.

9.1 A.1 Foundational Integrality Constraint

The integrality of the topological charge is not an artificially imposed quantization condition, but a rigid fact enforced by the fundamental homomorphism property of the exponential map on the complex plane.

Let $z_1, z_2 \in \mathbb{C}$. Then:

$$\exp(z_1) = \exp(z_2) \iff z_1 - z_2 = 2k\pi i, \quad k \in \mathbb{Z}$$

This property ensures that for any characteristic class defined by complex phase winding, the charge value must lie strictly in the discrete additive group $\mathbb{Z}$.

Framework Corollary: On the automorphic line bundle, the space of the bare first Chern class c_1^{bare} is strictly restricted to $\mathbb{Z}$, rather than the continuous real field $\mathbb{R}$. This integer skeleton is a necessary algebraic consequence of the weight k of the automorphic forms.

9.2 A.2 Rigid Constraints on the Running Scalar Decomposition

The decomposition $\mathbf{C}(\mu) = c_1^{\text{bare}} + \delta c_1(\mu)$ in Axiom III is subject to the following irreducibility conditions:

1. **Expectation Value Redefinition**: $\mathbf{C}(\mu)$ in Case B characterizes the "non-commutativity-induced effective curvature expectation value", rather than the strict global Chern class. The global topological charge c_1^{bare} is strictly non-deformable continuously.
2. **Restoration Criterion**: When the system returns to Case A (global recursive rigidity), $\delta c_1(\mu)$ must collapse to an integer value so that $\mathbf{C}(\mu) \equiv 0 \pmod{\mathbb{Z}}$, i.e., the running correction is completely absorbed into the integer renormalization without leaving a continuous tail.

9.3 A.3 Cutoff Boundary and Admissible Extensions

Let Λ be the cutoff scale of any class space:

- **Regular Boundary**: When the cutoff tends to infinity ($\Lambda \to \infty$), bare integrality is strictly restored, corresponding to strictly compact spectrum.
- **Discrete Boundary**: At finite cutoff ($\Lambda < \infty$), the effective topological charge k_{eff} may be formally expressed as $k_{\text{eff}} \sim k/f(\Lambda)$. Integrality is then elevated to a congruence constraint on the cutoff parameter:

$$\Lambda \cdot k_{\text{eff}} \equiv 0 \quad (\text{mod integer period})$$

9.4 A.4 Quantitative Criterion for the Imaginary Spectral Dimension and Integrality Breakdown

When the system attempts to break bare integrality on the extended boundary, the imaginary part d_y of the spectral dimension in Axiom IV must satisfy a precise renormalization group equation to absorb the phase shift.

Let the effective phase winding at boundary cutoff Λ be $\Theta(\Lambda) = 2\pi c_1^{\text{bare}} + \phi(\Lambda)$. If $\phi(\Lambda)$ is non-zero (fractional shift), then the imaginary spectral dimension d_y is strictly equal to:

$$d_y = \frac{1}{2\pi} \frac{d}{d \ln \Lambda} \arg\left(\phi(\Lambda)\right)$$

Judgement Theorem: $d_y \neq 0$ is the necessary and sufficient geometric condition that permits "integer overflow" or "renormalization group tunnelling" of the topological charge on the extended boundary. If $d_y = 0$, then $\phi(\Lambda)$ decays along the flow, fractional shifts are forbidden, and the system must return to the discrete spectrum of Case A.

9.5 Core Conclusion of the Framework

Quantum non-commutativity, continuous spectra, and gauge-field topological symmetry breaking are all pure algebraic consequences of the loss of global ordinal recursive linear dependence of automorphic classes within the NBG framework. The theory requires no externally imposed Planck constant, measurement hypothesis, or semi-classical approximation.

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